Rachel R. Sharp

Science Communicator & Neuroscientist

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Profile

I am a neuroscientist and science writer who learns unfamiliar science quickly and explains it clearly. As a writer, **most recently for *American Scientist***, I translate dense research for a broad readership, and as an editor and translation coordinator I have helped other writers do the same. My PhD integrated genetics, psychiatry, and population health to ask how genetic and social risk combine in bipolar disorder (**first-author paper in *HGG Advances*, 2026**); before that, as bioinformatic analyst for 20+ labs, I delivered clear, interpretable results to collaborators across a range of topics. I also believe science is grounded in the people who do it: I've mentored everyone from middle schoolers to professors, and as president of the Neurosciences Student Executive Committee during my PhD I led program reform based on feedback from 100+ students and staff. **I completed my PhD in September 2026, am relocating to Copenhagen in December, and am seeking roles in science communication, research outreach, and program coordination.**

Education

PhD in Neuroscience | University of North Carolina at Chapel Hill 2026
Certificate in Bioinformatics and Computational Biology • NSF Graduate Research Fellow

BS in Psychology | University of Oklahoma 2019
Summa Cum Laude (4.0 GPA) • Minor: Criminology • National Merit Scholar

Selected Writing

"Finding Humanity in Health Data" | *American Scientist* 2025
"When Practice Makes Perfect: The Neurons Behind Skilled Movement" | *The Pipettepen* 2024
"The Need for Diversity in Large-Scale Genetic Studies" | *The Pipettepen* 2023

Science Communication Experience

Author | *American Scientist*, Macroscope 2025
- Interviewed Dr. Oliver Bear Don't Walk IV about building intersectionality into medical informatics and wrote "Finding Humanity in Health Data", bringing that research to a general science readership
- Developed the piece during science communication training with ComSciCon-Triangle and took it through the magazine's editorial process to publication

Contributing Author | *BrainPost* 2023–2025
- Distilled peer-reviewed research, on topics such as neuroplasticity and brain-computer interfaces, into short, digestible articles for a non-specialist readership

Writer & Editor | *The Pipettepen* 2022–2025
- Wrote science articles for general readers, including "The Need for Diversity in Large-Scale Genetic Studies" (2023) and "When Practice Makes Perfect: The Neurons Behind Skilled Movement" (2024)
- Served as an editor, revising articles for message, structure, accuracy, and clarity

Blog Scheduler, Translation Coordinator, and Editor | *NC DNA Day Blog* 2022–2023
- Built and ran the blog's Spanish translation program, which raised translated output from 3 articles over two years to 17 articles in a single semester
- Managed the editorial calendar for writers, editors, and translators, keeping posts on schedule

Writing Assistant | OU Honors College 2018–2019
- Taught writing to 25+ Honors College students, coaching them on persuasion, clarity, and structure

Leadership and Outreach

President | UNC Neurosciences Student Executive Committee 2022–2025
- Represented 100+ graduate students in regular meetings with the program directors and their administrative staff, gathering feedback from junior students and implementing it as tangible program changes
- Orchestrated changes across the training program, including an overhauled qualifying exam, rewritten Annual Progress Review reports, anonymous reporting channels, and first-year check-ins, and built community through the annual retreat and events such as the chili cook-off

Session Moderator and Abstract Reviewer | American Society of Human Genetics (ASHG) 2024
- Reviewed and scored annual meeting abstracts; moderated the session "Phenomenal PheWAS"

Organizer | Human Functional Regulatory Genomics Group 2023–2025
- Ran monthly trainee meetings on functional genomics: set topics, recruited speakers, and led discussion

Mentor and Science Communicator | WinSPIRE UNC, Culbreth STEAM Day, and other programs 2022–2024
- Developed and led workshops on science and science communication for middle and high school students; mentored students through the WinSPIRE summer research program and Letters to a Pre-Scientist; taught R to 40 students as a volunteer instructor, UNC School of Medicine

Research Experience

Graduate Research Assistant | University of North Carolina at Chapel Hill (PI: Dr. Michael Love) 2021–2026
- Conceived, designed, and led a study of genetic and social risk for bipolar disorder in 115K *All of Us* participants, performing all analyses and orchestrating collaboration among clinicians, geneticists, biostatisticians, and population health researchers at UNC and Duke through first-author publication
  - Integrated whole-genome sequencing, electronic health records, and participant surveys to compare polygenic risk with six social risk measures across ancestry groups
  - Found that social risk contributed considerable explanatory power beyond genetic risk alone, with high social burden conferring substantial risk even among individuals in the lowest deciles of polygenic liability
  - Advocated for holistic models of psychiatric risk that integrate genetic liability with social and environmental context, showing that precision psychiatry frameworks that omit social risk provide an incomplete and less equitable picture of individual-level risk
- Coordinated a shared analysis pipeline across five internationally funded centers of the Impact of Genomic Variation on Function (IGVF) Consortium and led the cross-center meetings, setting analytical standards for a multi-institutional consortium
- Awarded NSF Graduate Research Fellowship ($111,000 stipend over 3 years), funding this research

Bioinformatic Analyst | OU Health Sciences Center Core Facilities (PI: Dr. Kenneth Jones) 2020–2021
- Partnered with 20+ Principal Investigators across oncology, immunology, vision, and metabolic disease, learning each new field quickly enough to deliver results and contributing to 6 co-authored papers (*PLOS Pathogens*, *Endocrinology*, *Frontiers in Immunology*, and others)
- Wrote custom R, Python, and Bash analyses of single-cell and bulk sequencing data and produced the figures and written results used in papers and grant applications

Lab Manager | OU Visual Neuroscience Lab (PI: Dr. Michael Wenger) 2018–2019
- Managed and trained 20+ research assistants to run EEG studies of iron deficiency and cognition

Selected Publications and Presentations

- **Sharp, R.R.**, Hysong, M.R., Mealer, R.G., Raffield, L.M., Glover, L., Love, M.I. (2026). Considering social risk alongside genetic risk for bipolar disorder in the All of Us Research Program. *Human Genetics and Genomics Advances* 7(4), 100665. doi.org/10.1016/j.xhgg.2026.100665
- 8 co-authored and consortium publications, including *Nature* (2024, IGVF Consortium), *Nature Genetics* (2025), *PLOS Pathogens* (2023), and *Endocrinology* (2021)
- 2 oral and 6 poster presentations, most recently “More Than the Sum of Our Genes: How Social Risk Shapes the Development of Bipolar Disorder” at the Festival of Genomics (London, 2026); also ASHG (Denver, 2024) and Genomics of Brain Disorders (Wellcome Genome Campus, UK, 2023)

Selected Honors

Featured Author, Inside HGGA Interview Series | *HGG Advances* 2026
National Science Foundation Graduate Research Fellowship 2023
Genomics of Brain Disorders Conference Bursary Travel Award 2023
Best Poster Award (2nd place) | *IGVF Consortium Annual Meeting* 2022
UNC Department of Neuroscience T32 Training Grant 2022
Big Woman on Campus Award | Campus-wide scholarship, leadership, and service 2019
Dean’s Award for Outstanding Honors College Leadership 2019

Skills

Communication: feature and plain-language writing, interviewing, editing, editorial planning, public speaking
Outreach and Organization: workshop design, event and program planning, stakeholder engagement, mentoring
Research: psychiatric genetics, polygenic risk, EHR phenotyping, social determinants of health; R, Python, Bash

Professional Development

ComSciCon-Triangle science communication workshop 2025
Science Communicators of North Carolina, member 2024–2025